Missing Eigenvectors

Consider $\boldsymbol{y}' = \boldsymbol{A}_\varepsilon \boldsymbol{y}$ with $\varepsilon \neq 0$.

$$\boldsymbol{A}_\varepsilon = \begin{pmatrix} \lambda & 1 \\ 0 & \lambda + \varepsilon \end{pmatrix}$$

The distinct eigenvalues give two independent solutions.

$$\lambda_1 = \lambda \qquad \boldsymbol{V}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \boldsymbol{y}_{1,\varepsilon}(t) = e^{\lambda t}\boldsymbol{V}_1$$

$$\lambda_2 = \lambda + \varepsilon \quad \boldsymbol{V}_2 = \begin{pmatrix} 1 \\ \varepsilon \end{pmatrix} \quad \boldsymbol{y}_{2,\varepsilon}(t) = e^{(\lambda+\varepsilon)t}\boldsymbol{V}_2$$

As $\varepsilon \to 0$, $\boldsymbol{V}_2 \to \boldsymbol{V}_1$ and both solutions approach $e^{\lambda t}\boldsymbol{V}_1$.

A Second Solution in the Limit

For $\varepsilon \neq 0$, the scaled difference is also a solution.

$$\frac{\boldsymbol{y}_{2,\varepsilon}(t) - \boldsymbol{y}_{1,\varepsilon}(t)}{\varepsilon} = e^{\lambda t} \begin{pmatrix} (e^{\varepsilon t} - 1)/\varepsilon \\ e^{\varepsilon t} \end{pmatrix} \xrightarrow{\varepsilon \to 0} e^{\lambda t} \begin{pmatrix} t \\ 1 \end{pmatrix}$$

The first entry inside the vector tends to t by L'Hôpital's rule.

For $\boldsymbol{W} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$, $(\boldsymbol{A}_0 - \lambda \boldsymbol{I})\boldsymbol{W} = \boldsymbol{V}_1$, so

$$\boldsymbol{y}_2(t) = e^{\lambda t}(t\boldsymbol{V}_1 + \boldsymbol{W})$$

solves $\boldsymbol{y}' = \boldsymbol{A}_0\boldsymbol{y}$. Its second entry is $e^{\lambda t}$, while $e^{\lambda t}\boldsymbol{V}_1$ has second entry zero.

Reduction of Order

The trick is to let the coefficients of $\boldsymbol{V}_1$ and $\boldsymbol{W}$ vary. For the limiting system $\boldsymbol{y}' = \boldsymbol{A}_0\boldsymbol{y}$, set

$$\boldsymbol{y}(t) = e^{\lambda t}\big(a(t)\boldsymbol{V}_1 + b(t)\boldsymbol{W}\big)$$

Substitute and cancel the terms from differentiating $e^{\lambda t}$. Since $(\boldsymbol{A}_0 - \lambda\boldsymbol{I})\boldsymbol{V}_1 = 0$ and $(\boldsymbol{A}_0 - \lambda\boldsymbol{I})\boldsymbol{W} = \boldsymbol{V}_1$, we get

$$a'(t)\boldsymbol{V}_1 + b'(t)\boldsymbol{W} = b(t)\boldsymbol{V}_1 \qquad \Longrightarrow \qquad a'(t) = b(t), \quad b'(t) = 0$$

Choose $b(t) = 1$ and $a(0) = 0$ to get $a(t) = t$. This gives the same second solution as the limit.

$$\boldsymbol{y}_2(t) = e^{\lambda t}(t\boldsymbol{V}_1 + \boldsymbol{W})$$

A Third Approach: The Taylor Series

For any square matrix $\boldsymbol{A}$, the shift $\boldsymbol{N} = \boldsymbol{A} - \lambda_1 \boldsymbol{I}$ commutes with $\lambda_1 \boldsymbol{I}$. The binomial theorem therefore lets us factor the exponential series:

$$e^{t\boldsymbol{A}} = \underbrace{\left(1 + \lambda_1 t + \frac{(\lambda_1 t)^2}{2!} + \cdots\right)}_{e^{\lambda_1 t}} \left(\boldsymbol{I} + t\boldsymbol{N} + \frac{t^2 \boldsymbol{N}^2}{2!} + \cdots\right).$$

For our limiting matrix, $\lambda_1 = \lambda$ and $\boldsymbol{N} = \boldsymbol{A}_0 - \lambda_1 \boldsymbol{I} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$, so $\boldsymbol{N}^2 = 0$ and the second series stops after the linear term.

$$e^{t\boldsymbol{A}_0} = e^{\lambda_1 t}(\boldsymbol{I} + t\boldsymbol{N}) = e^{\lambda_1 t} \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}.$$

Applying this matrix to $\boldsymbol{W}$ gives $e^{\lambda_1 t}(\boldsymbol{W} + t\boldsymbol{V}_1)$, the same second solution.

A Fourth Approach: Expand the Reduced Unknown

Set $\boldsymbol{y}(t) = e^{\lambda_1 t}\boldsymbol{u}(t)$ for the limiting system. Substituting and canceling the exponential gives

$$\boldsymbol{u}' = \boldsymbol{N}\boldsymbol{u}, \qquad \boldsymbol{N} = \boldsymbol{A}_0 - \lambda_1 \boldsymbol{I}, \qquad \boldsymbol{u}(0) = \boldsymbol{W}.$$

Write $\boldsymbol{u}(t) = \sum_{k=0}^{\infty} t^k \boldsymbol{u}_k$. Matching powers of t gives

$$(k+1)\boldsymbol{u}_{k+1} = \boldsymbol{N}\boldsymbol{u}_k, \qquad \boldsymbol{u}_0 = \boldsymbol{W}, \quad \boldsymbol{u}_1 = \boldsymbol{N}\boldsymbol{W} = \boldsymbol{V}_1, \quad \boldsymbol{u}_2 = \tfrac{1}{2}\boldsymbol{N}\boldsymbol{V}_1 = 0.$$

The recurrence makes every higher coefficient zero. Therefore

$$\boldsymbol{u}(t) = \boldsymbol{W} + t\boldsymbol{V}_1, \qquad \boldsymbol{y}_2(t) = e^{\lambda_1 t}\big(\boldsymbol{W} + t\boldsymbol{V}_1\big).$$